Software Testing

User Acceptance Testing, Regression Testing & STLC

1 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) is the final phase of testing where **end users or clients** verify whether the software meets the **business requirements** and is ready for production use.

1.1 Why UAT is Performed

- To validate real-world business workflows
- To ensure customer satisfaction
- To confirm the system is production-ready

1.2 When UAT is Performed

UAT is performed **after system testing** and **before final deployment**.

1.3 Who Performs UAT

- End users
- Clients or customers
- Product owners
- Business analysts

1.4 Types of UAT

- Business Acceptance Testing
- Alpha Testing
- Beta Testing
- Regulatory Acceptance Testing

1.5 Example of UAT

Online Shopping System

- User logs in
- Adds product to cart
- Makes payment

If all steps work correctly, the system is accepted.

1.6 Advantages and Limitations

- **Advantages:** Ensures business correctness, reduces post-release issues
- **Limitations:** Time-consuming, depends on user availability

2 Regression Testing

Regression Testing is a type of software testing conducted to confirm that a recent change, upgrade, or bug fix in the application has **not adversely affected the existing functionalities**. It ensures that previously working features continue to function correctly after modifications.

2.1 Why Regression Testing is Crucial

A regression testing approach is required to evaluate the overall working of the application after it undergoes changes. Since software components are often interdependent, a modification in one module can unintentionally affect other modules.

Key Reasons for Performing Regression Testing

- **Identifying Regression Defects**
  - Regression testing helps detect unintended defects introduced due to new code changes.
  - It ensures that newly added functionality does not interfere with existing features.
  - It also helps identify defects in both newly pushed code and previously stable modules.
- **Ensuring Software Stability**
  - Verifies that the existing functionality remains intact after enhancements or fixes.
  - Detects unexpected behavior that may impact user experience.
  - Helps maintain consistent system performance.
- **Mitigating Risks**
  - Identifies potential risks caused by system changes at an early stage.

  - Prevents system failures, performance degradation, and business disruptions.
  - Enhances overall reliability and user confidence in the application.

2.2 When Regression Testing is Performed

Regression testing is performed **whenever changes are made** to the software application.

- After adding a new feature
- After fixing defects or bugs
- After code refactoring
- Before every major release

2.3 Who Performs Regression Testing

- QA engineers
- Software testers
- Automated testing tools (for repeated execution)

2.4 Types of Regression Testing

- **Unit Regression Testing** – testing individual modules
- **Partial Regression Testing** – testing affected areas only
- **Complete Regression Testing** – testing the entire application

2.5 Example of Regression Testing

Scenario: E-commerce Website
After fixing a bug in the checkout module:

- Login functionality is tested
- Cart operations are verified
- Payment gateway behavior is re-tested

This ensures that the fix has not negatively impacted other parts of the system.

2.6 Advantages and Limitations of Regression Testing

- **Advantages**
  - Ensures long-term software stability

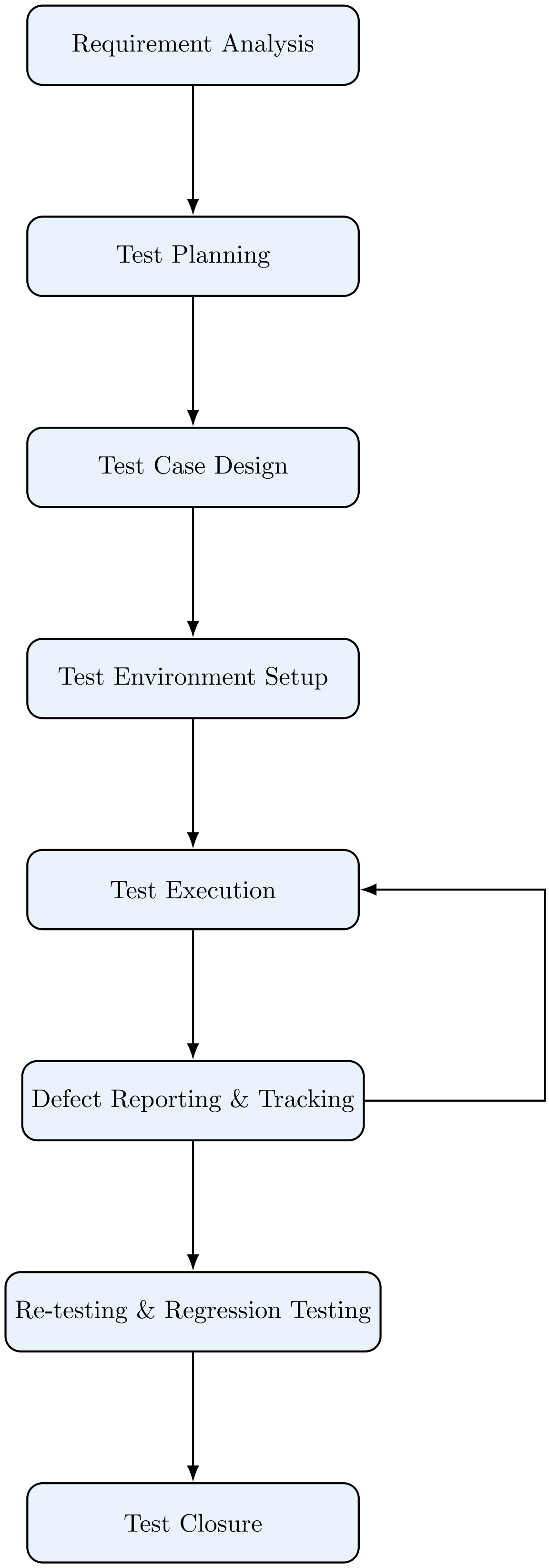
  - Detects unexpected side effects
  - Highly suitable for automation

- **Limitations**
  - Can be time-consuming for large systems
  - Requires maintenance of test cases

3 Comparison: UAT vs Regression Testing

Aspect	**UAT**	**Regression Testing**
Performed By	End users / Clients	QA / Testers
Focus	Business requirements	Existing functionality
When Performed	Before release	After any change
Objective	User satisfaction	Software stability
Automation	Mostly manual	Mostly automated

4 Software Testing Life Cycle (STLC)



5 Exam-Oriented Key Takeaways

- UAT validates **business requirements**
- Regression Testing validates **software stability**
- UAT is user-driven, Regression is tester-driven
- STLC is an iterative process